Simple Case of Graham and Powell (2012)

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Nurdaulet Menglibayev

1 one unit

Start with one unit and an explicit model. Also applies to N units just take N=T. For this single i, suppose

$$Y_{it} - W_{it}\delta_0 = X_{it}\beta_i + \varepsilon_{it}, \qquad t = 1, \dots, T,$$

where β_i is fixed and δ_0 is known. Use the example

$$X_{it} = S_i Z_{it}, \qquad S_i \sim U[0, b], \qquad Z_{it} \overset{\text{iid}}{\sim} N(0, 1).$$

Let the errors be independent of the regressors and have dentity cov (as in example) also for now assume normality. Creates simple classic OLS case.

Regression is

$$\boxed{b_i(X) - \beta_i = \frac{\sum_{t=1}^{T} X_{it}\varepsilon_{it}}{\sum_{t=1}^{T} X_{it}^2}.}$$

Now calculate its conditional squared error explicitly:

$$\begin{aligned} E[(b_i(X) - \beta_i)^2 \mid X_i] &= \frac{E\left[\left(\sum_t X_{it}\varepsilon_{it}\right)^2 \mid X_i\right]}{\left(\sum_t X_{it}^2\right)^2} \\ &= \frac{\sum_t \sum_s X_{it}X_{is}E(\varepsilon_{it}\varepsilon_{is} \mid X_i)}{\left(\sum_t X_{it}^2\right)^2}. \end{aligned}$$

Thus,

$$\boxed{\text{Given the observed } X_i, \quad \operatorname{Var}(b_i(X_i) \mid X_i) = (X_i'X_i)^{-1}.}$$

Because the same S_i multiplies every period,

$$\begin{aligned} \frac{1}{T}X_i'X_i &= S_i^2 \frac{1}{T}\sum_t Z_{it}^2 \\ &\longrightarrow S_i^2 \quad \text{almost surely.} \end{aligned}$$

More periods average the Z_{it}^2. They do not average the persistent S_i^2. Consequently,

$$\left(\frac{1}{T}X_i'X_i\right)^{-1} \longrightarrow \frac{1}{S_i^2},$$

a random limit across possible unit scales. For each fixed $s^2 > 0$,

$$\boxed{\sqrt{T}\,[b_i(X) - \beta_i] \mid S_i^2 = s^2 \xrightarrow{d} N(0, 1/s^2).}$$

For the unconditional single-unit limit, write $G \sim N(0,1)$, independent of S_i. Integrating the conditional distributions gives

$$\boxed{\sqrt{T}[b_i(X_i) - \beta_i] \xrightarrow{d} \frac{G}{S_i}.}$$

$$E\left[\left(\frac{G}{S_i}\right)^2\right] = E[G^2]E[S_i^{-2}] = \frac{1}{b}\int_0^b \frac{ds}{s^2} = \infty.$$

So the single-unit limiting mixture has infinite second moment. Also,

$$E\left[\left|\frac{G}{S_i}\right|\right] = \frac{\sqrt{2/\pi}}{b}\int_0^b \frac{ds}{s} = \infty.$$

Mean does not exist either.

1.1 Why the event $S_i \le k/\sqrt{T}$ dominates

Fix $k > 0$. For all sufficiently large T,

$$\boxed{P\left(S_i \le \frac{k}{\sqrt{T}}\right) = \frac{k}{b\sqrt{T}}.}$$

This is a shrinking event for one unit's distribution.

Normality gives the distributional representation

$$b_i(X_i) - \beta_i \overset{d}{=} \frac{G}{\sqrt{T}\,S_i\sqrt{\frac{1}{T}\sum_t Z_{it}^2}},$$

where $G \sim N(0,1)$ is independent of S_i and the Z_{it}. Since $T^{-1}\sum_t Z_{it}^2 \to 1$ almost surely, the scale $S_i = k/\sqrt{T}$ gives

$$b_i(X_i) - \beta_i \mid S_i = k/\sqrt{T} \xrightarrow{d} \frac{G}{k}.$$

Consequently,

$$\liminf_{T\to\infty} P\left(|b_i(X_i)-\beta_i|>1 \,\middle|\, S_i \le \frac{k}{\sqrt{T}}\right) \ge P(|G|>k) > 0.$$

Combining this with the probability of the shrinking event,

$$\boxed{\liminf_{T\to\infty} \sqrt{T}\, P\left(|b_i(X_i)-\beta_i|>1,\ S_i \le \frac{k}{\sqrt{T}}\right) \ge \frac{k}{b} P(|G|>k) > 0.}$$

On this event, the squared normalized error $T[b_i(X_i)-\beta_i]^2$ exceeds T. Its size therefore grows faster than the event's probability shrinks. Even truncating the squared normalized error at T gives

$$\begin{aligned} &E\left[\min\left\{T[b_i(X_i)-\beta_i]^2, T\right\}\right] \\ &\qquad \ge T\, P\left(|b_i(X_i)-\beta_i|>1,\ S_i \le \frac{k}{\sqrt{T}}\right). \end{aligned}$$

Hence,

$$\boxed{\liminf_{T\to\infty} \frac{E\left[\min\left\{T[b_i(X_i)-\beta_i]^2, T\right\}\right]}{\sqrt{T}} \ge \frac{k}{b} P(|G|>k) > 0.}$$

These finite, truncated expectations already diverge.

This does not contradict consistency: $b_i(X_i) \to \beta_i$ in probability. Conditional asymptotic normality for each fixed $S_i = s > 0$ remains valid.